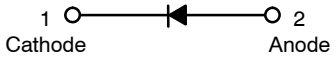
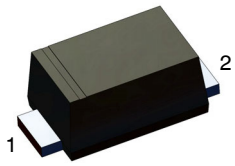
# Surface Mount Fast Recovery Rectifiers

## 0.8 A, 50 V - 1000 V

# RS1AFA, NRVHPRS1AFA Series




**SOD−123FA**
**CASE 425AB**

(Color Band Denotes Cathode)

### Features

- Glass Passivated Chip Junction
- Fast Switching for High Efficiency
- High Surge Capacity
- Low Forward Voltage: 1.3 V Maximum
- UL Flammability 94V−0 Classification
- MSL 1 per J−STD−020
- RoHS Compliant / Green Molding Compound
- Industrial Device Qualified per AEC−Q101 Standards
- NRVHP Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable

*See authorized use policy

### ORDERING INFORMATION

See detailed ordering, marking and shipping information in the package dimensions section on page 4 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 4.

**Table 1. ABSOLUTE MAXIMUM RATINGS** Values are at $T_A$ = 25°C unless otherwise noted.

| Symbol | Parameter | Value | | | | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | RS1 AFA | RS1 BFA | RS1 DFA | RS1 GFA | RS1 JFA | RS1 KFA | RS1 MFA | |
| $V_{RRM}$ | Repetitive Peak Reverse Voltage | 50 | 100 | 200 | 400 | 600 | 800 | 1000 | V |
| $V_{RMS}$ | RMS Reverse Voltage | 35 | 70 | 140 | 280 | 420 | 560 | 700 | V |
| $V_R$ | DC Blocking Voltage | 50 | 100 | 200 | 400 | 600 | 800 | 1000 | V |
| $I_{F(AV)}$ | Average Forward Rectified Current | 0.8 | | | | | | | A |
| $I_{FSM}$ | Peak Forward Surge Current: 8.3 ms Single Half Sine−Wave Superimposed on Rated Load | 30 | | | | | | | A |
| $T_J$ | Operating Junction Temperature Range | −55 to +150 | | | | | | | °C |
| $T_{STG}$ | Storage Temperature Range | −55 to +150 | | | | | | | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

**Table 2. THERMAL CHARACTERISTICS** (Note 1)

| Symbol | Parameter | Value | Unit |
|---|---|---|---|
| $\Psi_{JL}$ | Typical Thermal Characteristics, Junction−to−Lead | 32 | °C/W |
| $R_{\theta JA}$ | Typical Thermal Resistance, Junction−to−Ambient | 105 | °C/W |

1. Device mounted on 5 mm x 5 mm Cu pad PCB.

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**Table 3. ELECTRICAL CHARACTERISTICS** Values are at $T_A$ = 25°C unless otherwise noted.

| Symbol | Parameter | Conditions | | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $V_F$ | Instantaneous Forward Voltage (Note 2) | $I_F$ = 0.8 A | | | | 1.3 | V |
| $I_R$ | Reverse Current at Rated $V_R$ | $T_J$ = 25°C | | | | 5 | μA |
| | | $T_J$ = 125°C | | | | 50 | |
| $C_J$ | Junction Capacitance | $V_R$ = 4 V, f = 1 MHz | | | 10 | | pF |
| $T_{rr}$ | Reverse Recovery Time | $I_F$ = 0.5 A, $I_R$ = 1 A, $I_{rr}$ = 0.25 A | RS1AFA, RS1BFA, RS1DFA | | | 150 | ns |
| | | | RS1GFA, RS1JFA | | | 250 | |
| | | | RS1KFA, RS1MFA | | | 500 | |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Pulse test with PW = 300 μs, 1% duty cycle



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## TYPICAL PERFORMANCE CHARACTERISTICS

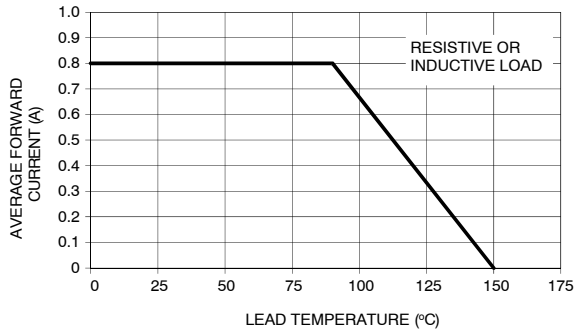



**Figure 1. Forward Current Derating Curve**

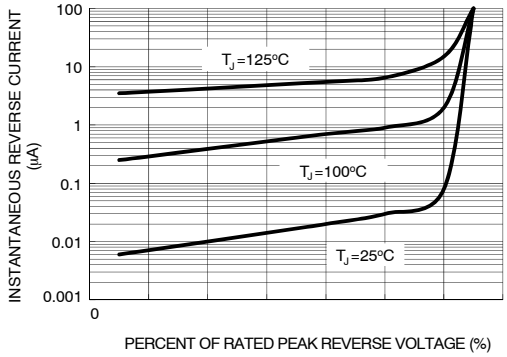



**Figure 2. Typical Reverse Characteristics**

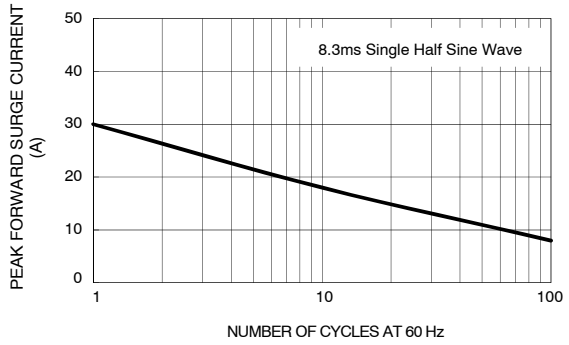



**Figure 3. Maximum Non−Repetitive Forward Surge Current**

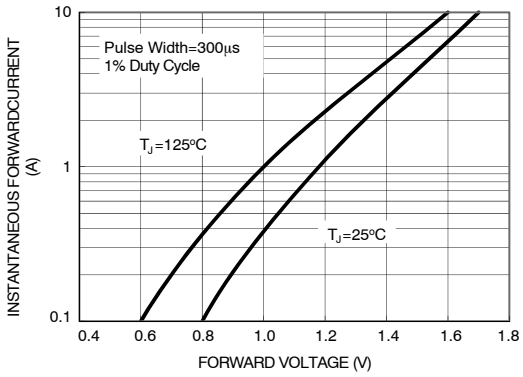



**Figure 4. Typical Instantaneous Forward Characteristics**

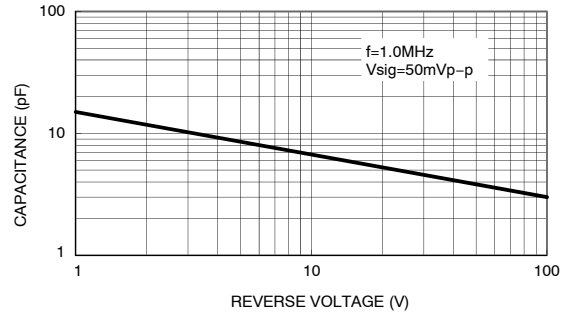



**Figure 5. Typical Junction Capacitance**

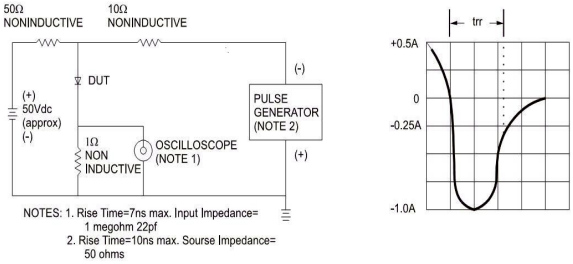



**Figure 6. Reverse Recovery Time Characteristic and Test Circuit Diagram**



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## ORDERING INFORMATION

| Device | Top Mark | Package | Shipping† |
|---|---|---|---|
| RS1AFA | RAL | SOD−123FA | Tape and Reel |
| RS1BFA | RBL | SOD−123FA | Tape and Reel |
| RS1DFA | RDL | SOD−123FA | Tape and Reel |
| RS1GFA | RGL | SOD−123FA | Tape and Reel |
| RS1JFA | RJL | SOD−123FA | Tape and Reel |
| RS1KFA | RKL | SOD−123FA | Tape and Reel |
| RS1MFA | RML | SOD−123FA | Tape and Reel |
| NRVHPRS1AFA | RAL | SOD−123FA | Tape and Reel |
| NRVHPRS1JFA | RJL | SOD−123FA | Tape and Reel |
| NRVHPRS1KFA | RKL | SOD−123FA | Tape and Reel |
| NRVHPRS1MFA | RML | SOD−123FA | Tape and Reel |

**DISCONTINUED** (Note 3)

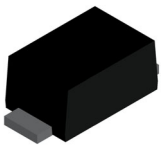
| | | | |
|---|---|---|---|
| NRVHPRS1BFA | RBL | SOD−123FA | Tape and Reel |
| NRVHPRS1DFA | RDL | SOD−123FA | Tape and Reel |
| NRVHPRS1GFA | RGL | SOD−123FA | Tape and Reel |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

3. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

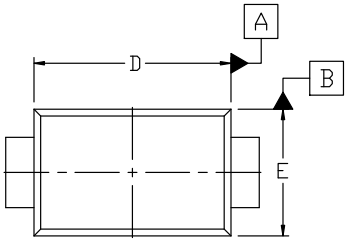


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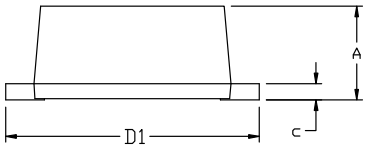


**SOD−123FA**
CASE 425AB
ISSUE A

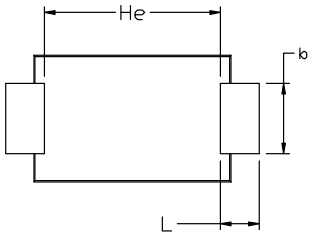
DATE 11 AUG 2022




TOP VIEW




FRONT VIEW

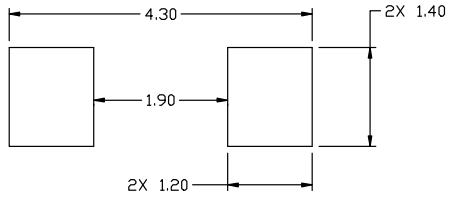



BOTTOM VIEW

NOTES:

1. NO INDUSTRY STANDARD APPLIES TO THIS PACKAGE.
2. ALL DIMENSIONS ARE IN MILLIMETERS.
3. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND THE BAR PROTRUSIONS.

| DIM | MILLIMETERS | | |
|---|---|---|---|
| | MIN. | NOM. | MAX. |
| A | 1.23 | 1.33 | 1.43 |
| b | 0.80 | 1.00 | 1.20 |
| c | 0.16 | 0.23 | 0.30 |
| D | 2.70 | 2.80 | 2.90 |
| D1 | 3.40 | 3.60 | 3.80 |
| E | 1.70 | 1.80 | 1.90 |
| He | 2.45 | --- | 2.60 |
| L | 0.35 | 0.60 | 0.85 |




RECOMMENDED
MOUNTING FOOTPRINT*

* For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

| | | | |
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